

WEIDA CHEN

Game Engineer

weidac490@gmail.com | github.com/Weida758 | linkedin.com/in/weida-chen-800457347

TECHNICAL SKILLS

Languages: C++, C#, Python

Engine: Unity **Tools:** Git, GitHub, Visual Studio, Rider

Concepts: Gameplay Systems, OOP/OOD, State Machines, Object Pooling, ScriptableObjects, Data Structures & Algorithms, Linear Algebra

EDUCATION

University of California, Irvine

Expected June 2028

Double Major: Computer Science; Game Design and Interactive Media

GPA: 3.85

Relevant Coursework: Game Programming Series, Game Design, Linear Algebra, Programming in C/C++; Computer Networks, Game Engineering, Data Structures & Algorithms

PROJECTS

Get Me Out | Unity, C#

2026

Game Programmer | UCI Zotjam | **2nd Judge Choice, 4th Audience Choice**

- Implemented an idle/movement finite state machine integrating Unity Input System actions with card-triggered player abilities.
- Implemented turn, jump, dash, and gravity inversion using Rigidbody2D velocity, forces, and gravity control.
- Built a LinkedList-based deck with pre-game card selection/retrieval, last-in-first-out ability execution, and event-driven UI updates.

Knight-mare Survival | Unity, C#

2026

Game Programmer | School Project | Survival Game

- Engineered weighted enemy spawning with timed unlocks, active-enemy caps, and time-based scaling of wave size and enemy stats.
- Built prefab-keyed object pools for enemies, XP gems, and VFX, prewarming and reusing instances to reduce repeated instantiation and destruction.
- Built a modular weapon and upgrade pipeline using ScriptableObject data, weapon levels, global damage bonuses, and upgrade options.
- Connected XP, level-up, and death events to UI, queueing pending level-ups and pausing gameplay during upgrade selection.

CPU Path Tracer | C++

Tutorial Project | Based on Ray Tracing in One Weekend

- Implemented a CPU path tracer with ray-sphere intersections, recursive ray scattering, and diffuse, metallic, and dielectric materials.
- Implemented multisample antialiasing, gamma correction, and a configurable perspective camera with depth of field.